Fairness Formulation Decision Note

Fair and Capacity-Constrained Bipartite Matching | Draft for team review before Week 3

Status: Provisional recommendation only. The team should approve, revise, or reject it after reviewing the literature.

1. Recommended formulation

Primary candidate: max-min utility among maximum-cardinality allocations. First maximize the number of assigned students. Among allocations with that maximum assignment count, maximize the minimum normalized utility received by any assigned student. This prioritizes avoiding very poor outcomes without sacrificing the maximum achievable assignment count.

Let $x_{ij} \in \{0,1\}$ indicate that student i is assigned to project j; c_j is project capacity; $u_{ij} \in [0,1]$ is normalized utility; and A is the set of assigned students.

Stage 1: $K^* = \max \sum_{i,j} x_{ij}$, subject to eligibility, one-project-per-student, and project-capacity constraints.

Stage 2: maximize $z = \min_{i \in A} \sum_j u_{ij} x_{ij}$, subject to $\sum_{i,j} x_{ij} = K^*$ and the same feasibility constraints.

Utility convention: convert preference ranks to a documented normalized scale (for example, rank 1 = 1.0, rank 2 = 0.75, rank 3 = 0.5, rank 4 = 0.25). The team must approve the scale and use it consistently. Evaluate assigned students and report the number unassigned separately.

2. Why this candidate?

It fits the student-project model, needs no demographic/group labels, and can be measured from preferences and final assignments. It can be compared against the maximum-cardinality, maximum-weight, and min-cost max-flow baselines. Its limitation is that results depend on the utility scale and it may sacrifice total utility to improve the worst assigned outcome.

3. Secondary reporting metric

Report average normalized utility and/or average preference rank among assigned students, alongside assignment count, minimum utility, and runtime. These are reporting metrics; do not optimize them as a second objective unless the team deliberately adopts a multi-objective formulation.

4. Shortlist decisions

Approach	Decision	Reason
Max-min utility	Primary candidate	Directly focuses on the worst assigned utility; measurable within project scope.
Envy-freeness	Do not select as primary yet	Requires a precise envy definition; may conflict with capacities and preferences or be infeasible.
Group parity	Defer unless justified groups are available	Needs approved group labels; these are optional in the current project model.
Nash social welfare	Keep as alternative	Balances utilities but needs careful zero handling and potentially harder optimization.

5. Input and implementation requirements

Required inputs: student and project IDs, eligible pairs, project capacities, preference ranks or normalized utilities, and final assignments. Group labels are **not required** for this candidate. If group fairness is later selected, define the attribute, justification, and missing-data policy.

Keep binary assignment variables, $\sum_j x_{ij} \le 1$, $\sum_i x_{ij} \le c_j$, and $x_{ij} = 0$ for ineligible pairs. Stage 1 can use matching or flow; Stage 2 may require ILP or a suitable optimization formulation. Complexity depends on the exact utility representation and solver. Verify feasibility and report runtime; do not claim NP-hardness without a source or proof for the exact formulation.

6. Evaluation and definition of done

Compare algorithms on identical input instances. Report assigned count, minimum normalized utility among assigned students, average utility or preference rank, runtime, and capacity/eligibility violations. Keep unassigned count visible so fairness cannot be improved by silently dropping students.

Decision requested: Before Week 3, the team should approve, revise, or reject max-min utility as the primary candidate; confirm secondary reporting metrics; and document the decision. Until then, this remains a recommendation, not a finalized metric.